

Collin Cranston

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EDUCATION

University of California, San Diego

La Jolla, California

PhD in Mathematics, with Specialization in Statistics

June 2026

- Thesis: *Random Matrix Theory for Linearized Neural Networks*
- Research: deep learning theory, random matrix theory, neural networks, overparameterization theory
- GPA: 3.79

MS in Statistics

March 2023

- Original project: Generalized Cross Validation (GCV) estimator in high-dimensional ridge regression.
- GPA: 3.71

University of California, Los Angeles

Los Angeles, California

BS in Mathematics

June 2019

- Highest departmental honors
- GPA 3.58

PUBLICATIONS AND PREPRINTS

- **NTK Regression under dual power-law model: Deterministic Equivalents via SDE and PDE Methods**
C. Cranston; Z. Wang; T. Kemp *Accepted, NeurIPS (Main) 2026*
- **Eigen-Spike Emergence and Quadratic Equivalents for Conjugate Kernels on Nonlinearly Separable Data**
C. Cranston*; Z. Wang*; T. Kemp; M. Mahoney. [arxiv:2605.29669](https://arxiv.org/abs/2605.29669) *Conference on Learning Theory (COLT), 2026*

RELEVANT EXPERIENCE

Cotality

Data Science Intern

June – August 2026

- Built a production-grade multi-agent system (Google ADK) that drafts insurance regulatory filings for Cotality's predictive models by grounding each answer in prior filings via RAG (Vertex AI) and pulling supporting evidence from internal documents and code repositories through MCP servers.
- Designed the agent as a deterministic graph workflow with human-in-the-loop review and an LLM-as-a-judge validator, ensuring auditability and reliability for a compliance-sensitive regulatory process.
- Demonstrated that the agent would cut filing time and staffing requirements substantially; addressing the main barrier to scaling further carrier adoption of Cotality's models.

Handshake AI

AI Trainer- Mathematics Expert

June-August 2025

- Worked with several top AI labs to create and evaluate original tasks in mathematics that were used to assess and fine-tune performance of large language models (LLMs) in mathematical reasoning.

SLMath Summer Machine Learning Graduate School

Teaching Assistant

July 2023

- Designed and led daily problem-solving sessions for graduate students at the SLMath-funded Summer School on Machine Learning, writing exercises to accompany lectures on deep learning theory, multidimensional scaling, and reinforcement learning.

PROJECTS

- Deep Convolutional Neural Network (CNN) for semantic segmentation task on PASCAL VOC-2012 dataset.
- Recurrent Neural Network (RNN) and LSTM model for NLP task on Shakespearean text generation.
- Brain tumor localization and classification via several CNN vision models architectures, e.g. YOLO and R-CNN.
- Two-stage meta-labeling framework for tactical asset allocation of across equities, bonds, and commodities. Applied Random Matrix Theory eigenvalue filter for feature selection.

ADDITIONAL

- **Programming Languages:**
 - Python (PyTorch, TensorFlow, NumPy, Pandas, Keras, Matplotlib, Scikit-learn, SciPy, etc.), SQL, MATLAB, R
- **Honors and Awards:**
 - James B. Ax Fellowship, UCSD 2020
 - Highest Departmental Honors, UCLA 2019